

- 19** Thermometers M_1 and M_2 are placed inside an evacuated enclosure with walls maintained at temperature T . The thermometers are identical except that the bulb of M_1 is blackened.

Which of the following equations is true if T_1 and T_2 are the temperatures indicated by M_1 and M_2 respectively after thermal equilibrium has been established?

- A** $T_1 > T_2 > T$ **B** $T_1 = T_2 = T$ **C** $T_2 > T_1 = T$ **D** $T_1 > T_2 = T$

19	B	At thermal equilibrium, the temperature is the same for the thermometers and the walls.
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- 20** A fixed mass of ideal gas at constant pressure occupies a volume V . The gas undergoes a